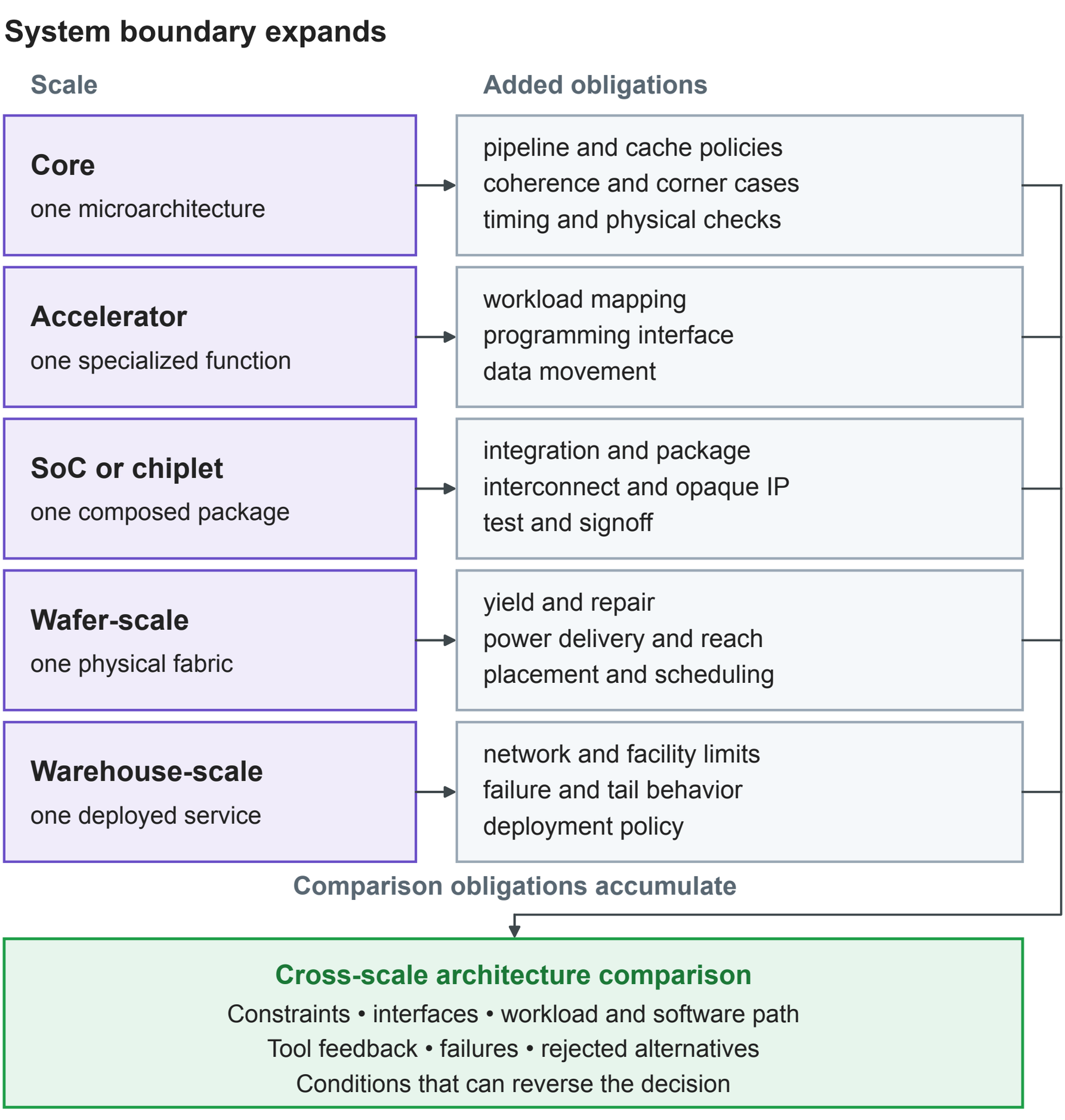
System boundary expands
Scale
Added obligations
Core
one microarchitecture
pipeline and cache policies
coherence and corner cases
timing and physical checks
Accelerator
one specialized function
workload mapping
programming interface
data movement
SoC or chiplet
one composed package
integration and package
interconnect and opaque IP
test and signoff
Wafer-scale
one physical fabric
yield and repair
power delivery and reach
placement and scheduling
Warehouse-scale
one deployed service
network and facility limits
failure and tail behavior
deployment policy
Comparison obligations accumulate
Cross-scale architecture comparison
Constraints • interfaces • workload and software path
Tool feedback • failures • rejected alternatives
Conditions that can reverse the decision